

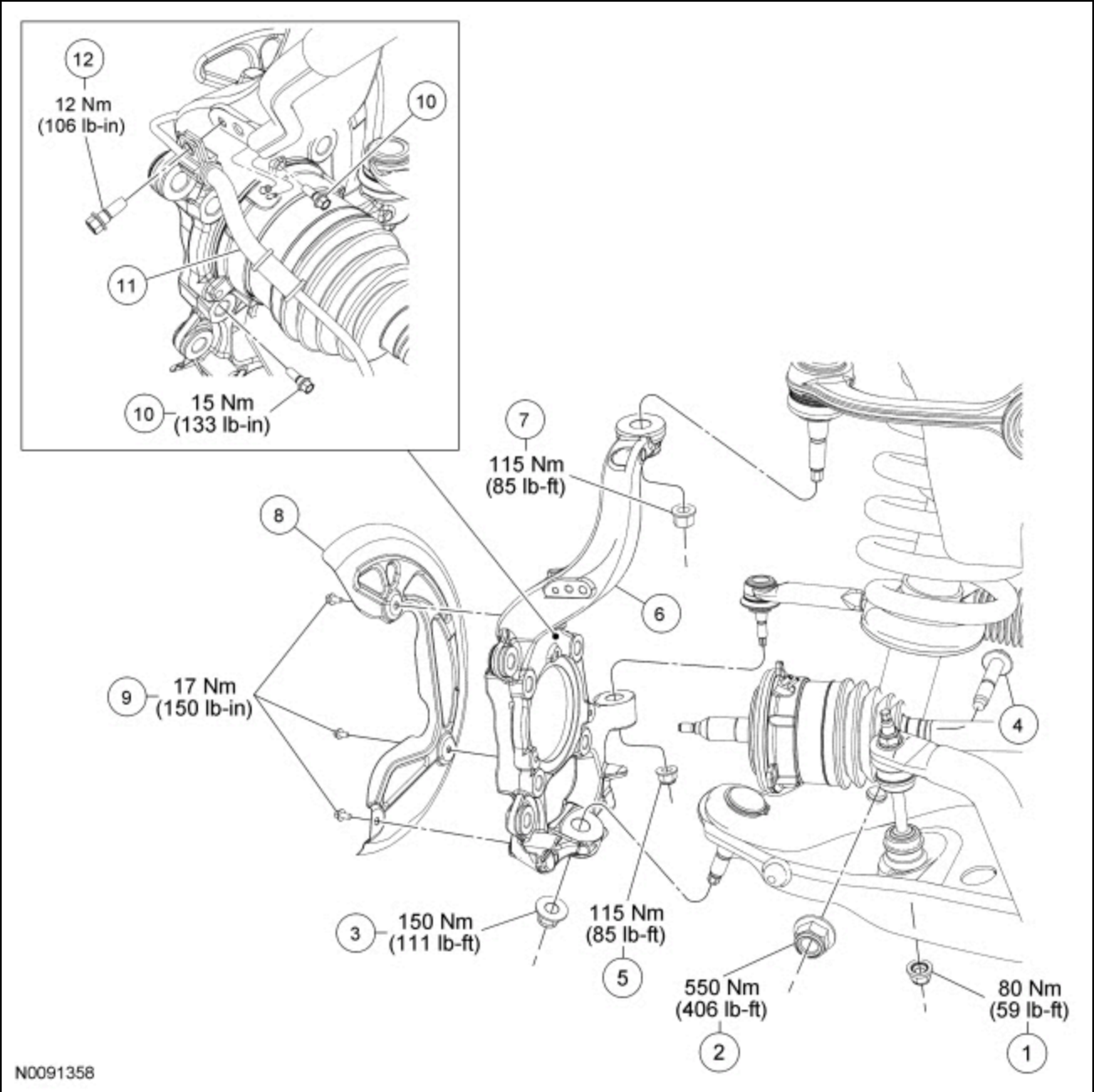
Wheel Knuckle

 [Ford Ecat Electronic Parts Catalog](#)

Special Tool(s)

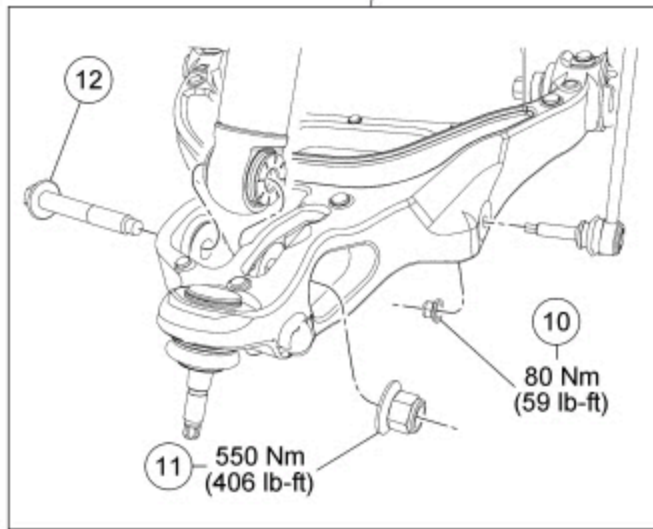
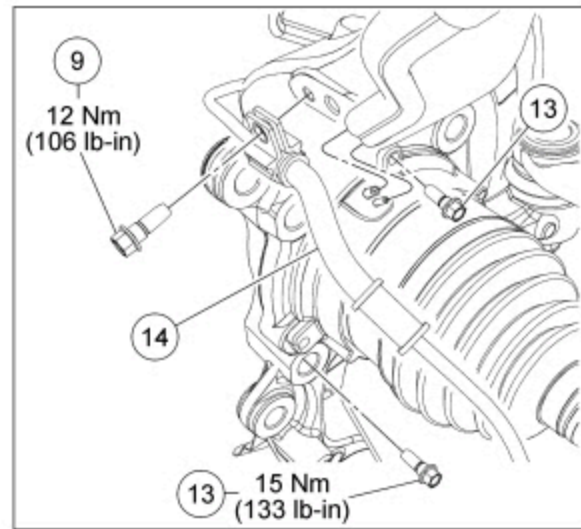
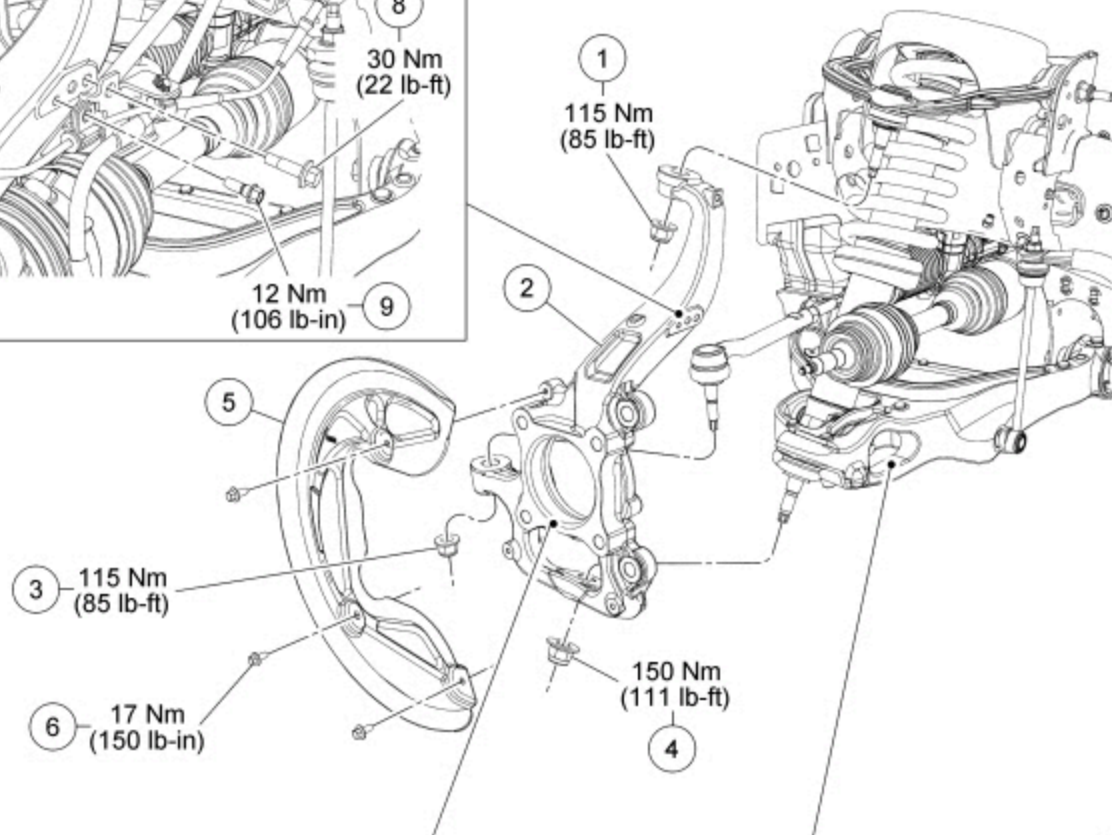
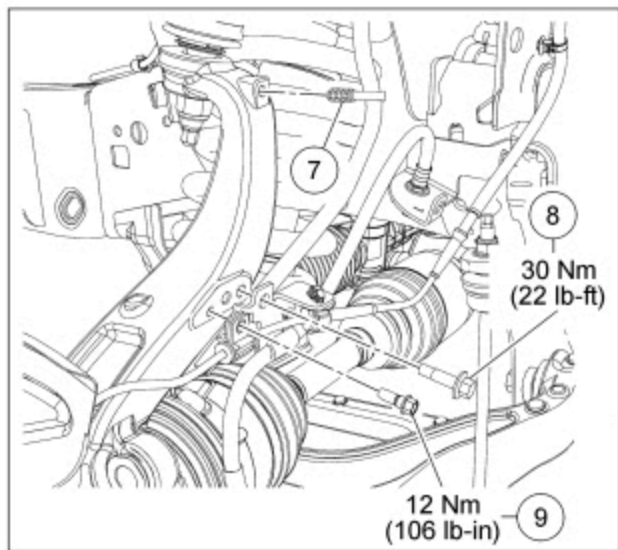
 ST2945-A	Separator, Ball Joint 204-592
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All Except SVT Raptor



Item	Part Number	Description
1	N807144 N807144	Stabilizer bar link lower nut
2	W520217 W520217	Shock absorber lower nut
3	W707254 W707254	Lower ball joint nut
4	W712682 W712682	Shock absorber lower bolt
5	W520215 W520215	Tie-rod end nut
6	3K186 3K186 LH/ 3K185 RH LH/ 3K185 RH	Wheel knuckle
7	W520215 W520215	Upper ball joint nut
8	2K005 2K005 LH/ 2K004 RH LH/ 2K004 RH	Brake disc shield
9	W505251 W505251	Brake disc shield bolts (3 required)
10	W710279 W710279	Integrated Wheel End (IWE) bolts (3 required)
11	—	Wheel speed sensor harness bracket (part of 2C204)
12	W710279 W710279	Wheel speed sensor harness bracket bolt

SVT Raptor



N0105877

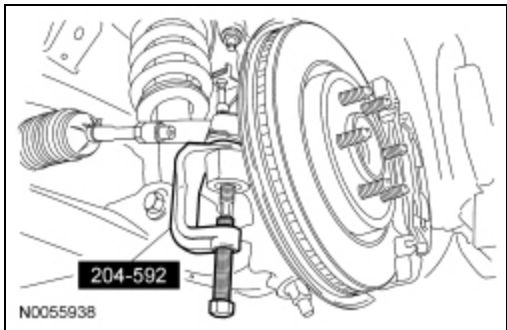
Item	Part Number	Description
1	W520215 W520215	Upper ball joint nut
2	3K186 3K186 LH/ 3K185 RH LH/ 3K185 RH	Wheel knuckle
3	W520215 W520215	Tie-rod end nut
4	W707254 W707254	Lower ball joint nut
5	2K005 2K005 LH/ 2K004 RH LH/ 2K004 RH	Brake disc shield
6	W505251 W505251	Brake disc shield bolts (3 required)

7	—	Integrated Wheel End (IWE) vacuum hose push-pin retainer
8	W506423 W506423	Brake hose bracket bolt
9	W710279 W710279	Wheel speed sensor harness bracket bolt
10	W807144 W807144	Stabilizer bar link lower nut
11	W707019 W707019	Shock absorber lower nut
12	W712104 W712104	Shock absorber lower bolt
13	W710279 W710279	IWE bolts (3 required)
14	—	Wheel speed sensor harness bracket (part of 2C204)

Removal

NOTICE: Suspension fasteners are critical parts because they affect performance of vital components and systems and their failure may result in major service expense. New parts must be installed with the same part numbers or equivalent part, if replacement is necessary. Do not use a replacement part of lesser quality or substitute design. Torque values must be used as specified during reassembly to make sure of correct retention of these parts.

1. Remove the wheel bearing and wheel hub. For additional information, refer to Wheel Bearing and Wheel Hub in this section.
2. Remove and discard the tie-rod end nut.
3. Using the Ball Joint Separator, separate the tie-rod end from the wheel knuckle.



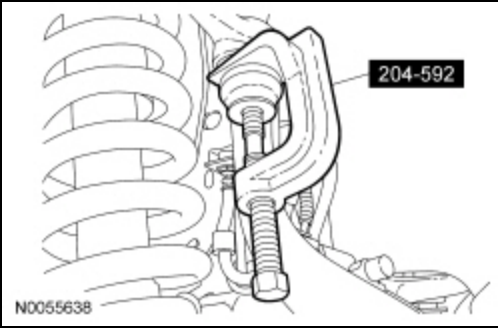
4. Remove the wheel speed sensor harness bracket bolt and position the harness aside.
5. Remove the brake hose bracket bolt and position aside.
6. **NOTE:** Use the hex-holding feature to prevent the stud from turning while removing the nut.

Remove and discard the stabilizer bar link lower nut.

7. Remove and discard the shock absorber lower nut and bolt.
8. Remove the 3 Integrated Wheel End (IWE) bolts.
9. Remove and discard the upper ball joint nut.

10. **NOTE:** Be sure not to damage the ball joint boot when installing the Ball Joint Separator.

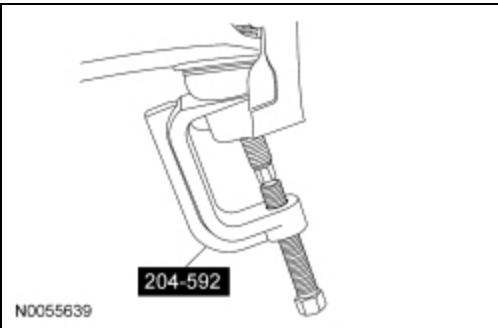
Using the Ball Joint Separator, separate the upper ball joint from the wheel knuckle.



11. Remove and discard the lower ball joint nut.

12. **NOTICE:** Make sure to use the Ball Joint Separator when separating the lower ball joint from the knuckle to prevent damage to the ball joint boot.

Using the Ball Joint Separator, separate the lower ball joint from the wheel knuckle and remove the wheel knuckle.



13. If necessary, remove the 3 brake disc shield bolts and remove the brake disc shield.

Installation

NOTICE: Do not tighten the lower shock nut until the installation procedure is complete and the weight of the vehicle is resting on the wheel and tire assemblies or incorrect clamp load and bushing damage may occur.

NOTE: When servicing the shock absorber lower nut, take care not to damage the stabilizer link.

1. If necessary, install the brake disc shield and 3 brake disc shield bolts.
 - Tighten to 17 Nm (150 lb-in).
2. Position the lower ball joint into the wheel knuckle and install the new nut.
 - Tighten to 150 Nm (111 lb-ft).
3. Position the upper ball joint and install the new nut.
 - Tighten to 115 Nm (85 lb-ft).
4. Position the IWE and install the 3 IWE bolts.
 - Tighten to 15 Nm (133 lb-in).
5. **NOTE:** Do not tighten the nut at this time.

Loosely install the new shock absorber lower nut and bolt.

6. **NOTE:** Use the hex-holding feature to prevent the stud from turning while installing the nut.

Install the new stabilizer bar link lower nut.

- Tighten to 80 Nm (59 lb-ft).

7. Position the wheel speed sensor harness bracket and install the bolt.

- Tighten to 12 Nm (106 lb-in).

8. Position the brake hose bracket and install the bolt.
 - Tighten to 30 Nm (22 lb-ft).
9. Install the new tie-rod end nut.
 - Tighten to 115 Nm (85 lb-ft).
10. Install the wheel bearing and wheel hub. For additional information, refer to Wheel Bearing and Wheel Hub in this section.
11. With the weight of the vehicle on the wheel and tire, tighten the shock absorber lower nut to 550 Nm (406 lb-ft).